The Oxford Handbook of the Foundations and Regulation of Generative AI

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CHAPTER

Introduction to the Foundations and Regulation of Generative AI

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Abstract

This chapter introduces *The Oxford Handbook of the Foundations and Regulation of Generative AI*, outlining the key themes and questions surrounding the technical development, regulatory governance, and societal implications of generative AI. The chapter highlights the historical context of generative AI, distinguishes it from traditional AI, and explores its diverse applications across multiple domains, including text, images, music, and scientific discovery. The discussion critically assesses whether generative AI represents a paradigm shift or a temporary hype. Furthermore, the chapter extensively surveys both emerging and established regulatory frameworks, including the EU AI Act, the General Data Protection Regulation, privacy and personality rights, and copyright, as well as global legal responses. We conclude that, for now, the 'Old Guard' of legal frameworks regulates generative AI more tightly and effectively than the 'Newcomers', but that may change as the new laws fully kick in. The chapter concludes by mapping the structure of the handbook.

Keywords: Generative AI, artificial intelligence regulation, large language models, AI governance, algorithmic transparency, AI ethics, foundation models, AI Act, GDPR, copyright

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I. A Prompt Start

Since 30 November 2022, most people associate generative AI with a chatbot able to produce coherent and sometimes even elegant sentences but making the occasional factual mistake. This popular account, however, assumes both too much and too little. It assumes too much because, historically, generative AI technologies have been around for a longer period,[1] and, substantively, the sentences large language models (LLMs) produce are, in essence, nothing more than an educated guess about what many users of the Internet—or authors of the documents included in the specific training set available to the model—would have written in a particular context. LLMs cannot 'think' or 'reason.' They predict, with increasing accuracy, the next token of a string of text based on patterns observed in their training data. While the output often seems coherent and knowledgeable, the current type of generative AI text engines cannot get rid of actual mistakes or wrongful attributions (often labeled 'hallucinations'), among other vulnerabilities.[2]

Simultaneously, the equation of generative AI and chatbots assumes too little, as generative AI is fundamentally about ordering and calculating sequential information—which goes much beyond text. It extends not only to pixels in images,[3] frames in videos,[4] and tones in music,[5] but also to DNA sequences,[6] protein structures,[7] mathematical objects,[8] and entire processes governed by agentic systems.[9]

This volume sets out to explore these broader implications of generative AI, particularly as they intersect with the foundational questions of ethics, governance, and societal impact. The *Oxford Handbook on the Foundations and Regulation of Generative AI* charts the theoretical underpinnings of generative AI, examining the ways in which these systems operate, their epistemic limitations, and their implications for our understanding of governance, applications, and regional as well as international repercussions. By bringing together contributions from leading thinkers across disciplines, this handbook not only clarifies the current state of the field but also anticipates the emerging technical, legal, and societal questions that will shape the future of generative AI.

In this introduction, we outline core elements of that discussion. Section II of this introduction traces the evolution from traditional AI to generative AI, providing context for its rapid rise. Section III evaluates whether generative AI represents more than a transient hype. Section IV surveys key regulatory developments and challenges shaping this technology. Section V considers the future trajectory of generative AI regulation. Finally, Section VI provides an overview of the Handbook, and details its contributions to the broader discourse.

II. From Traditional to Generative AI

While there is no universally accepted definition of generative AI, a shared understanding of its core features can be identified. 'Generative AI' refers to a subset of artificial intelligence designed to create new content—such as text, images, music, or code, and more.[10] These systems are regularly trained on vast datasets and often operate with diverse input formats, including text, audio, and images, enabling flexible, multimodal interactions and content generation.

By contrast, what we refer to as traditional AI systems (also called predictive AI, non-generative AI or discriminative modeling[11]) are primarily built to perform *specific* tasks or solve problems using predefined rules, statistical models, or machine learning algorithms. These systems typically process structured inputs and generate structured outputs, such as classifications or predictions—essentially numbers, not text, images, videos, or other communicative content. Examples of traditional AI include fraud detection in financial systems and medical diagnostics.

Generative AI models are often based on transformer-based architectures,[12] particularly foundation models[13] and large language models (LLMs),[14] such as a Generative Pre-Trained Transformer (GPT),[15] as well as image generators based on diffusion models[16] (used for systems such as DALL·E or Stable Diffusion). Unlike traditional AI, these models generate new data that not only resemble their training data but also include communicative content, such as text or code, images, videos, or music. This extends their use to creative outputs. Generative AI systems simulate human-like creativity and assist in tasks such as writing (e.g. generating online content or programming code), creating art and design, composing music, or synthesizing realistic speech. As explored in this book, their applications span a wide range, covering creative tasks and innovation, health, education, cybersecurity, arbitration, agriculture, and finance.

When comparing traditional AI and generative AI, it becomes clear that many challenges associated with traditional AI persist—and may even intensify—with generative AI. These include issues related to bias, data quality, transparency, sustainability, societal impact, and cybersecurity and systemic risks in critical sectors, which will be discussed in detail in the chapters of this book.

A distinguishing feature of generative AI systems is their 'heterogeneity': the term encompasses a wide array of systems with diverse architectures and capabilities, unified by their common feature that they create new content, not by the underlying technology. Additionally, some systems, particularly LLMs, have a well-documented tendency to produce 'hallucinations'—outputs that are false or misleading owing to a lack of connection to reality—as well as 'careless speech'—outputs that lack evidential grounding, references, or otherwise lack appropriate care for truthfulness or verifiability.[17] Another rather unique feature is the (controversial, see below) potential for so-called capability overhang, especially in foundation models, where the true capabilities of a system are claimed to surpass what its creators initially envisioned, and what its users are aware of, leading to unanticipated and undetected risks and opportunities.

Finally, generative AI is also a matter of scale: it consumes more data, compute and energy than any previous AI technologies.[18] As a consequence, it is uniquely expensive to train and run, both at the development and the inference level,[19] making return on investments particularly challenging, although distilling complex models may provide a pathway to achieve comparable performance with greater efficiency.[20]

III. Is Generative AI More Than Hype?

These idiosyncrasies in terms of projected risks and capabilities have led to vastly polarizing statements, often highlighting only one side of the coin and ignoring the other.

A. Hype Galore

On the one hand, the generative AI landscape is awash with evangelical proclamations that border on technological messianism. Tech leaders have been particularly effusive, with Bill Gates declaring that generative AI 'has the potential to change the world in ways that we can't even imagine',[21] while Sam Altman boldly predicts it will be 'the greatest force for economic empowerment' ever seen.[22] These grandiose statements reflect a widespread enthusiasm that sometimes overshadows critical analysis. To a certain extent, warnings about advanced AI systems posing an existential risk to humanity[23] have added fuel to the fire of AI hype.

However, not all voices are unequivocally optimistic. Saurajit Kanungo, president of CG Infinity, offers a more measured perspective, warning that AI is 'a huge high-risk bet' where 'a large percentage of your investment will likely go nowhere'.[24] The INSEAD survey also reveals a more nuanced corporate attitude, with excitement and concern roughly in balance.[25] And while, in mid-2023, 90% of data scientists surveyed at an AI conference believed the hype surrounding generative AI was beyond justified;[26] only 12% of business leaders feel generative AI has ‘fundamentally’ changed software development.[27] This suggests that while the potential is indeed substantial, the current reality does not (yet) match some of the more breathless rhetoric.

B. Promising Avenues

If the history of AI shows one thing, overblown expectations have been predictably followed by an AI winter. However, this is not an inevitability. Beyond all the hype, generative AI is indeed making a difference in the number of relevant scenarios, and it shows significant promise in empirical studies ranging from business analytics to coding and science.

A study conducted with Boston Consulting Group evaluated the performance implications of AI on complex, knowledge-intensive tasks. The experiment involved 758 consultants and found that those using AI completed tasks more quickly and with higher quality compared to those without AI access—in certain scenarios (e.g. generating at least 10 ideas for a new shoe targeting an underserved market or sport).[28] As in several other studies, the benefit was particularly strong for low-skilled workers.[29] Similar results were found for coding,[30] writing tasks,[31] and customer support.[32] A field experiment suggests that generative AI can be useful for providing unmonitored support services for information retrieval.[33] Another experiment explored generative AI’s influence on employee creativity in telemarketing, concluding that AI augmentation can enhance creative outputs—this time for highly skilled workers.[34] Similar results were reported in an MIT study where generative AI helped more experienced scientists most in discovering new viable and useful materials.[35] Both tasks required users to possess significant domain knowledge in order to assess and further develop the generative AI output, which likely favored higher-skilled individuals.[36]

In a more analytical task, however, in which spreadsheet data and statements from qualitative interviews needed to be jointly assessed and compared, groups using generative AI tools performed worse in the BCG study. Without extensive prompting, the AI tool was unable to extract subtle information from interviews needed to reverse the conclusion on what seemed to emanate from the spreadsheet data. Here, AI users were ‘falling asleep behind the wheel’[37]: they trusted AI too much in a task that lies beyond what the authors have dubbed its ‘jagged technological frontier’.[38] Hence, it is crucial to understand that generative AI can lead to performance benefits, but only under certain preconditions and in specific tasks that often involve test idea generation, simple data analysis, writing, or coding. If, on the other hand, generative AI is used blindly by workers or students without sufficient AI literacy, they will likely underperform in a variety of tasks beyond the frontier or in those which require specific prior knowledge to catch mistakes or careless speech.[39] In addition, studies suggest that even for tasks that are successfully completed, users of generative AI may forget the content more quickly than if they searched for and worked with the results themselves.[40] Hence, particularly environments that prioritize learning over fast performance should use generative AI with care.

Generative AI augmenting human performance in certain work tasks and under certain conditions has become a frequent topic of discussion in the recent literature. However, since generative AI works well with any type of sequential data, it also applies to sequences other than letters, such as DNA, molecular structures, and other objects of scientific discovery. It is not without reason that the 2024 Nobel Prize in Chemistry was awarded to researchers paving the ground for solving highly complex problems of protein folding with generative models, such as the AlphaFold family.[41] On the flipside, the use of generative AI may lead to even more fake science or output of low scientific quality.[42]

Used responsibly, generative AI may indeed spur on scientific discovery. A particularly promising trend lies in hybrid models, which aim to combine the information processing and productive capabilities of generative AI with the rigor of logical reasoning (e.g. neurosymbolic AI).[43] For example, DeepMind’s AlphaProof scored a silver medal at the 2024 International Mathematical Olympiad.[44] It includes Gemini, Google’s generative AI model, as an ‘informal model’ component, trained on mathematical proofs to reduce hallucinations and to phrase problems and answers. To rein it in, it is accompanied by a formal, code-based proof verifier. Similar approaches are starting to appear in other scientific disciplines in a trend that could ultimately fundamentally change the way research is conducted and researchers are trained.[45]

Such enhancements, the MIT study quoted above suggests, may come at a cost, however: the price might be less satisfaction of the researchers going about their tasks. The survey showed that 82% of the scientists in the experiment on material research reported being less satisfied with their work as a result of decreased utilization of their own creativity and skills.[46] Again, this shows that not only the quantity of work accomplished but also the quality of the work counts in human-machine teaming.[47]

Overall, this brief overview shows that empirical studies increasingly suggest generative AI can spark significant performance improvement across many sectors, including economic workflows, coding, and science. However, these early findings also indicate that the devil is in the details. A more nuanced understanding is necessary to unveil in which situations exactly generative AI is beneficial from an efficiency perspective, and in which it induces, on average, errors and technology overreliance detrimental to the desired outcomes.

C. Cautionary Tales

We do not, unfortunately, possess a crystal ball to chart the path for future development and adoption of generative AI. However, amid all the positive news, there are three key reasons for caution, stemming from the technology's social effects, technical limitations, and legal and regulatory challenges.

Concerning the first, as suggested above, generative AI, in its current version, has sparked significant controversy for making use of copyrighted works without proper remuneration of authors.[48] It is prone to deliver factually incorrect or wrongly referenced outcomes ('hallucinations' or 'careless speech'), which can exacerbate misinformation and disinformation online[49] and has been shown, in many instances, to produce results riddled with bias against protected groups, ranging from inadequate representation all the way to offensive and abusive content.[50] Those are serious and socially disruptive elements that are, to a large extent, baked into the current architecture of generative AI.

On the technical end, speculation ranges wildly concerning generative AI's future trajectory. One rather extreme projection suggests that AI training could scale up to 10^{29} floating-point operations (FLOPs) by 2030.[51] This would be three orders of magnitude larger than current frontier models, potentially enabling models far surpassing current capabilities. In line with these projections, Sam Altman speculated about 'Artificial General Intelligence' (AGI) and reiterated in January 2025 that, at OpenAI, 'we are now confident we know how to build AGI as we have traditionally understood it.'[52]

Others are less convinced. According to the Wall Street Journal, OpenAI's development of its next major model, GPT-5 (codenamed Orion), has encountered delays and escalating costs, indicating that the anticipated growth toward 'AGI' may face significant obstacles.[53] With (non-synthetic) training data finite and largely exhausted, future developments and breakthroughs might come from different architectures and using existing data in novel ways rather than scaling models through several orders of magnitude.

Moreover, recent studies highlight critical general issues in generative AI's development. Concerning the limitations of training data, models are plagued by the 'curse of recursion', meaning that training models on AI-generated data (a subset of 'synthetic data') leads to performance degradation, a phenomenon termed 'model collapse.'[54] Other findings reveal that state-of-the-art large language models can experience complete reasoning breakdowns when confronted with simple tasks, raising concerns about their reliability.[55] Additionally, it can be questioned whether the so-called emergent abilities observed in large language models (‘capability overhang’) are merely manifestations of in-context learning, which would challenge assumptions about undiscovered capabilities in current models.[56]

These findings underscore the necessity for cautious and critical approaches to the future development and deployment of generative AI technologies, and the need to challenge hyperbole and obvious marketing rhetoric, while acknowledging reliable evidence of the technology facilitating significant but gradual improvements in some areas and even astonishing breakthroughs in others.

IV. Key Regulatory Developments and Issues

The third and key factor for caution about the development and deployment of generative AI, next to social concerns and technological limitations, is the regulatory landscape that has seen dramatic shifts in the last years across many countries. While several new laws have been enacted, the majority of enforcement actions still rests on technology-neutral, ‘old’ regulation.

As a result, (generative) AI technologies increasingly face a fragmented regulatory landscape, characterized by significant cross-jurisdictional inconsistencies.[57] Divergent approaches to AI regulation, exemplified by the EU’s comprehensive AI Act and the more permissive frameworks adopted by the US and the UK, as well as the different but stringent regulations in China, and new AI-specific regulation in South Korea and envisaged in Canada and other nations, highlight fundamental differences in priorities, from AI safety and fundamental rights protection to innovation incentives. These discrepancies present formidable compliance challenges for multinational entities navigating multiple legal regimes, as they must reconcile varying obligations and standards while maintaining operational efficiency and technological competitiveness. Such conflicts underscore the inherent tension in balancing global AI governance with regional policy objectives and reflect the inevitability of regulatory clashes in an interconnected world of global generative AI rollout.

A. The ‘Newcomers’: AI Acts and Digital Regulation

The most important and salient addition to the regulatory landscape are several recent laws designed as comprehensive, AI-specific legislation, which include provisions on generative AI.

1. The EU AI Act

While China had taken the lead in formulating concrete legislative provisions aimed at generative AI licensing and oversight,[58] the EU passed the first comprehensive measure with its ambitious and widely applicable AI Act. The instrument applies to any AI system or model offered in the EU, and even those whose output is used in that area, endowing the AI Act with a global reach similar to other market regulations (such as competition law or the GDPR). Concerning generative AI, the AI Act introduces four key provisions[59]: transparency rules; provisions for all generative AI models; specific AI safety provisions for particularly powerful ones; and a far-reaching AI literacy requirement for any person dealing, in a professional capacity, with an AI model related to the EU. We take them up in turn.

First, according to Article 50 AI Act, any systems interacting with humans—such as chatbots—must be transparently identified as AI-enabled to break the illusion of a human counterpart. Furthermore, deployers (i.e. professional users) of generative AI must disclose deepfakes by marking the output as artificially generated or manipulated (‘generated by AI’). Note that the definition of deepfakes goes far beyond the rendering of persons to include seemingly authentic buildings, events, and recognizable landscapes (‘persons, objects, places, entities or events’, Art. 3(60)), with vast implications for marketing. Finally, providers must label generative AI output with state-of-the-art machine-readable techniques, such as watermarks. While these are increasingly available for audio, images and videos,[60] text watermarking is still in its infancy,[61] but may become state-of-the-art before the provision applies in August 2026.

Second, Article 53 AI Act lays down general rules for all general-purpose AI (GPAI) models, of which large generative AI models are the most important instantiation (Recital 99). It applies from August 2025 onward. On the one hand, providers of GPAI models must disclose a long list of technical items both to the European regulator, the AI Office within the European Commission, and to downstream deployers. This includes information on performance, training data, and even the energy consumed for model development.

On the other hand, Article 53 AI Act contains specific provisions that are relevant for copyright law and copyright holders. Still, the AI Act does not establish a new copyright regime for AI. Rather, it contains rules facilitating enforcement of existing EU law. Providers need to make available a detailed summary of training data sources, based on a template provided by the AI Office. Furthermore, and more importantly, GPAI model providers must 'put in place a policy to comply with Union law on copyright and related rights'. This has two main implications. First, the AI Act is commonly read as clarifying that a European copyright exception for text and data mining (in the DSM Directive) also covers AI training. This exception makes copyrighted material available for AI training without charge but introduces an opt-out right for rights holders in case the training pursues a commercial purpose. Second, Recital 106 AI Act suggests that, even if the AI model is trained outside of the EU, EU copyright law including particularly the opt-out right must be followed if the model is later used in the EU. Hence, even if US or Chinese copyright law does not require an opt-out right in a specific setting, training an AI model without respect for this EU-specific right in the US or China, respectively, may amount to a violation of the AI Act if the model is later used in the EU. This provision stands in contrast to the territoriality principle in international copyright law. However, it does not violate that principle as the AI Act does not make EU copyright law itself applicable in the US, China, or elsewhere outside of the EU. Nevertheless, if properly enforced, the provision marks a significant expansion of the reach of EU copyright law across the globe, via the bridge of the AI Act.

The third provision on generative AI in the AI Act is the technically most far-reaching one. Article 55 AI Act mandates that providers of general-purpose AI models with systemic risk must evaluate their models using standardized protocols, including adversarial testing, to identify and mitigate systemic risks. They are also required to assess and address potential risks at the Union level, report serious incidents to the AI Office and national authorities without undue delay, and ensure adequate cybersecurity for both the AI models and their physical infrastructure. However, these basic AI safety provisions apply only to a subgroup of GPAI models (with systemic risk). While the Commission can theoretically designate models falling into this category, the practically most important presumption holds that any model trained with more than 10^{25} FLOPs of compute constitutes a systemically risky GPAI model (Art. 51(2) AI Act). In January 2025, according to estimates, this comprised at least 17 models available on the market, such as GPT-4, GPT4o, Mistral Large and Large 2, Grok-2, Llama 3.1-405 B, Claude 3 Opus, Claude 3.5 Sonnet, and Gemini 1.0 Ultra as well as 1.0 and 1.5 Pro.[62]

It is important to note that deployers may become providers of generative AI models under the AI Act if they fine-tune or otherwise modify the original model (Recital 109). Moreover, using a GPAI model in scenarios defined by the AI Act as a high-risk domain (such as employment, credit scoring, health or life insurance, or certain educational use cases) also makes the deployer responsible for adhering to the high-risk rules of the AI Act (Article 25(1) AI Act). Hence, the provisions just described do not only apply to the handful of original GPAI providers (such as OpenAI, Anthropic, Microsoft, DeepMind or Mistral), but may easily extend to professional users.

Any deployer or provider within the scope of the AI Act, however, must also comply with the final and most organizationally far-reaching provision: Article 4. According to this norm, any entity providing or deploying an AI system or GPAI model must ensure that by February 2025, all staff and agents dealing with an AI system possess adequate AI literacy. This includes an understanding of the technical basis and limitations of AI as well as the main regulatory constraints and obligations (cf. Art. 3(56) AI Act). The provision covers millions of persons in a professional context, from C-level executives to employees, in any company, department or agency developing or using an AI model built for the EU market, or whose output is used in the EU—and it comes with a tight deadline for ensuring adequate AI literacy.

2. South Korea, Brazil, Canada, and the Vatican

South Korea has also made strides by enacting the AI Basic Act, becoming the second country globally to implement such legislation.[63] This law empowers the Minister of Science and ICT to establish and execute a national AI basic plan every three years. The Act focuses on transparency and AI safety, and mandates measures to ensure the reliability of AI, such as implementing watermarks on AI-generated content to combat issues like fake news, deepfakes, and violations of personal information and copyright.[64] Enforcement mechanisms empower the Ministry of Science and ICT to conduct investigations and issue corrective orders for violations. The AI Basic Act is set to take effect in January 2026, providing a one-year preparation period for stakeholders.

Brazil, in turn, has followed suit and, as of January 2025, is advancing a broad law inspired by the EU AI Act.[65] Established by Bill No. 2,338/2023, the legislation has been approved by the Federal Senate and is currently under consideration in the House of Representatives.[66] The measure introduces a risk-based framework requiring developers to classify AI systems, with excessive-risk systems prohibited and high-risk systems subject to specific compliance measures such as algorithmic impact assessments and human supervision. The Bill also establishes protections for individuals, including rights to privacy, non-discrimination, and human involvement in decisions, while prohibiting harmful uses of AI and seeking to promote innovation through copyright exceptions and experimental sandboxes.[67] It would take effect one year after its enactment.

In Canada,[68] the proposed Artificial Intelligence and Data Act (AIDA)[69] seeks to regulate international and interprovincial AI systems, focusing on mitigating risks of harm and bias. As of the time of writing this chapter, the law has not yet been enacted. AIDA introduces obligations for AI system providers and users, emphasizing transparency, accountability, and human rights protection. Similar to the AI Act, it takes a risk-based approach, focusing particularly on ‘high-impact AI systems’ that could significantly affect public safety.[70]

Notably, also the Vatican City State has introduced wide-ranging Guidelines on Artificial Intelligence, which have entered into force on 1 January 2025.[71] They range from high-level fundamental principles[72] to the prohibition of certain AI practices[73] and include principles on the use of AI in administrative procedures[74] and principles relating to the use of AI in judicial proceedings.[75] The Guidelines further provide for the establishment of a Commission on Artificial Intelligence, which is mandated with preparing implementing laws and regulations, as well as with ancillary activities, such as expressing opinions for AI trials and applications, monitoring activities and reporting on the impact of AI.[76]

3. DMA, DSA, and Cybersecurity Laws

Beyond AI-specific laws, several new pieces of digital legislation, particularly in the EU, also impact generative AI. One important puzzle piece is provided by the Digital Markets Act (DMA). In an effort to restore workable competition to the digital economy, it regulates so-called gatekeepers (i.e. platforms with particularly wide-ranging market power over certain central services in the digital economy). As of September 2024, the European Commission designated several major companies as gatekeepers, which largely coincide with the major generative AI providers, including: Alphabet, Amazon, Apple, ByteDance, Meta, and Microsoft.[77] The DMA not only prohibits data sharing between different services of gatekeepers, including for AI training, but also digs at the very heart of (generative) AI usage in the digital economy: ranking, which must be conducted in a fair, transparent and non-discriminatory manner.[78] This is particularly relevant for search engines and e-commerce rankings, both of which are increasingly supplemented by and integrated into generative AI tools. The DMA also contains access rights for business users which are meant to empower them to potentially compete with gatekeepers on their merits by tapping into part of their proprietary data source.

The Digital Services Act (DSA), in turn, is the EU's flagship instrument for platform regulation and mitigating online harms from hate speech, fake news and other problematic forms of content. It places significant emphasis on the protection of minors by mandating privacy, safety, and security measures (Article 28(1) DSA), which is a significant concern when generative AI meets attention-seeking platforms. The AI Act complements these protections by requiring providers of high-risk AI systems to consider potential adverse effects on minors and other vulnerable groups (Article 9(9) AI Act). Platforms categorized as Very Large Online Platforms (VLOPs) or Very Large Online Search Engines (VLOSEs), such as Facebook, Instagram, Snapchat, or Twitter/X, must conduct risk assessments addressing issues like bias, fake news, gender-based violence, but also harms to minors' physical and mental well-being (Article 34(1) DSA) and mitigate identified risks through appropriate measures (Article 35 DSA). VLOPs and VLOSEs are also subject to external audits (Article 37 DSA).[79]

This is particularly relevant for hybrid platforms increasingly integrating generative AI models into platform services. As mentioned, generative AI systems with systemic risks must adopt corresponding risk management strategies in line with Article 55 of the AI Act.[80] Additionally, the DSA requires VLOPs to monitor and manage harmful content, including AI AI-generated material, while all hosting platforms must maintain notice-and-action mechanisms to remove illegal content (Articles 16 and 34 DSA).

Concerning the particularly relevant field of cybersecurity—a crucial component of digital infrastructure in an age of increasing geostrategic implications of AI, geopolitical tensions, and state-sponsored advanced hacking attempts—the EU has recently enacted a string of new instruments (NIS 2 Directive, Cyber Resilience Act, and Cybersecurity Act), which also strengthen cybersecurity provisions for AI models,[81] not least via the implicit reference to these acts in Article 55 AI Act and the revised Product Liability Directive.[82]

4. US Approaches

Meanwhile, in the US, specific AI regulation consists of an intricate patchwork of federal initiatives and state level regulations.[83] In 2023, President Joseph Biden issued an 'Executive Order on the Safe, Secure, Trustworthy Development and Use of Artificial Intelligence',[84] which was aimed at establishing comprehensive oversight of AI development and deployment in federal agencies, focusing on safety, security, and trustworthiness.[85] This executive order was accompanied by the Office of Management and Budget (OMB) Memoranda 24-10, 'Advancing Governance, Innovation, and Risk Management for Agency Use of Artificial Intelligence', which provided specific guidance to federal agencies. However, in January 2025, on his first day in office, newly elected President Donald Trump rescinded EO 14100 and OMB Memoranda 24-10, and instead issued a new 'Executive Order on Removing Barriers to American Leadership in Artificial Intelligence'.[86] The EO states that '[i]t is the policy of the United States to sustain and enhance America's global AI dominance in order to promote human flourishing, economic competitiveness, and national security,' and it called for a 180-day review of all relevant rules and regulations.

At the same time, export controls on AI chips establishing a three-tiered system for international AI technology access (the US Framework for Artificial Intelligence Diffusion)[87,88] enacted in January 2025 also under former President Biden, were kept intact for the moment.

In the absence of specific federal AI regulations, several US states have initiated their own AI-related legislative measures, contributing to a complex and varied regulatory landscape.[89] California, for instance, has been proactive in this domain, introducing multiple bills aimed at addressing AI's implications across various sectors. Notably, the 'Safe and Secure Innovation for Frontier Artificial Intelligence Models Act' (SB 1047) sought to mandate safety evaluations for advanced AI models to mitigate potential risks associated with their deployment, but ultimately was not signed by Governor Newsom. Additionally, California has enacted laws targeting specific AI components and applications,[90] such as the prohibition of AI-generated deepfake content intended to deceive voters during elections and protections against deepfake nude imagery.[91] This legislation reflects the state's commitment to preserving election integrity and protecting citizens from AI-driven misinformation and harm.

In March 2024, Tennessee enacted the 'ELVIS Act,' becoming the first state to regulate AI simulations of image, voice, and likeness. This legislation addresses concerns over unauthorized use of an individual's likeness through AI-generated deepfakes, particularly in the entertainment industry.[92] Utah's 'Artificial Intelligence Policy Act' (SB 149), signed into law in March 2024, establishes liability for companies that fail to disclose their use of generative AI when required by state consumer protection laws.[93] It also creates the Office of Artificial Intelligence Policy and the Artificial Intelligence Learning Laboratory Program, aiming to oversee AI use and promote education in the field. At the municipal level, New York City implemented the Bias Audit Law (Local Law 144) in July 2023, requiring companies to conduct independent audits of their automated employment decision tools to prevent discrimination.[94] Colorado has enacted its own AI Act mandating transparency of AI tools interacting with consumers.[95] Effective 1 February 2026, it also requires that developers and deployers of high-risk AI systems exercise reasonable care to prevent algorithmic discrimination, particularly in consequential decisions affecting areas like education, employment, and healthcare.[96] Reasonable care is presumed to have been exercised if the developer or deployer complies with the specific provisions of the Act.

While addressing general concerns, these state- and city-level initiatives contribute to a patchwork of regulations that businesses must navigate even within the US—not to mention the even more complex tapestry in the global perspective.

B. The ‘Old Guard’: From Data Protection to Copyright

Next to these ‘newcomer’ laws, generative AI is also governed by existing, technology-neutral legal frameworks. This ‘Old Guard’ of laws exerts significant influence over the regulatory landscape, with data protection laws such as the General Data Protection Regulation (GDPR) in the European Union playing a pivotal role,[97] but also areas such as unfair competition, competition law, discrimination prohibitions, and copyright exerting a major influence.[98] Increasingly, liability, for example, product liability and defamation or libel law, will also play an important role.[99]

1. Data Protection Law

The European Data Protection Board (EDPB) has established a ChatGPT Taskforce to coordinate enforcement activities and ensure compliance with data protection standards. In May 2024, the EDPB’s ChatGPT Taskforce released a report detailing their efforts to oversee and guide the deployment of ChatGPT within the EU. The taskforce emphasized the importance of transparency, data minimization, and the protection of individuals’ rights in the context of AI-driven services.[100]

In December 2024, the EDPB adopted its much-anticipated Opinion 28/2024 on AI and the GDPR,[101] which provides guidance on when AI models can be considered anonymous, the applicability of legitimate interest as a legal basis for processing personal data in AI development and deployment, and the consequences of using unlawfully processed personal data in AI models. This opinion underscores the necessity for AI developers to ensure compliance with the GDPR throughout the AI lifecycle. Importantly, it rightly notes that generative AI models *themselves* may, depending on the circumstances, be considered personal data.[102] This has vast implications, as it may ultimately open the door for actions concerning the correction and erasure of entire models, as an ultima ratio. As one of us has discussed elsewhere, generative AI models can be likened to encrypted messages.[103] If decryption (i.e. the revelation of personal information via intentional attacks: model inversion or membership attacks) or luck (memorized data triggered by prompt), is reasonably likely, the model constitutes personal data under Art. 4(1), Recital 26 GDPR and the Breyer doctrine[104] of the CJEU. While this involves scrutiny in individual cases, it is not unreasonable to assume that, under some conditions, generative AI models will be considered personal data by regulators and courts, with the described consequences.

In fact, enforcement action has already started. In December 2024, Italy’s privacy regulator, Garante, imposed a €15 million fine on OpenAI for processing users’ personal data without an adequate legal basis, for lacking transparency in its ChatGPT application, and inadequate age controls for access of minors.[105] And just like it blocked ChatGPT temporarily in 2023, the Garante moved swiftly to block the app of the popular Chinese generative AI startup DeepSeek in Italy over questions surrounding transparency, the legal basis for processing and training, the protection of minors, and data protection by design.[106] Belgium equally opened an investigation, and other EU countries sent questions to the company.[107] Moreover, while the transfer chapter of the GDPR likely does not apply (as the data is transmitted from the DeepSeek API right to Chinese servers, and not via an EU ‘exporter’),[108] serious issues arise over the safeguards DeepSeek can offer, if any, concerning access by the Chinese government to its customer data. As the US TikTok case shows, the Chinese government has, apparently, rather unfettered access to data held by Chinese companies.[109]

Similar issues arise, however, with respect to US-based companies. In California, Microsoft’s LinkedIn was sued in January 2025 in a class action alleging the unauthorized disclosure of private messages to third parties for AI model training without user consent, alleging breach of California’s unfair competition law and the federal Stored Communications Act. This case underscores the necessity for organizations to potentially obtain explicit consent before utilizing personal data for AI training purposes,[110] even in the US.[111] Data protection extends, of course, beyond the GDPR and the US. In late January 2025, the South Korean regulator opened an investigation into DeepSeek over data protection concerns.[112] Again, this testifies to the global reach, and heavy impact, of data protection rules on the processing operations powering generative AI, both in training and inference.

2. Unfair Competition, Hallucinations, and Non-discrimination

In the US, the Federal Trade Commission (FTC) is also actively addressing potential unfair trade practices associated with AI technologies under federal law.[113] The FTC, and its EU counterparts in competition law, have also initiated investigations into collaborations between major generative AI companies, including partnerships like those between OpenAI and Microsoft, as well as AWS and Anthropic, to assess their impact on market competition.[114]

'Hallucinations', in turn, are tackled legally on multiple fronts. Privacy laws concerning personality rights, such as defamation and libel, are applicable to factually wrong generative AI outputs in both the EU and the US.[115] However, such data can also be challenged under the data accuracy principle of the GDPR (Art. 5(1)(d) GDPR), as another high-profile case started by privacy activist Max Schrems against OpenAI shows.[116] If hallucinations amount to 'misinformation', the provisions of the DSA will also offer alleviation (see Chapter IV DSA).

As one of us has described elsewhere, careless speech is a particularly subtle but important type of hallucination which poses unique regulatory challenges.[117] Careless speech consists of factual inaccuracies, misleading references and biased information that is produced by LLMs and distributed into the information ecosystem. It requires prior domain knowledge to detect and thus presents a different challenge than obvious hallucinations or misinformation and disinformation.[118] The scope of the challenge posed by careless speech remains murky as its frequency in major generative AI models is not yet well researched, but one prior study found that general-purpose LLMs hallucinate between 60–80% of the time when prompted with legal queries.[119]

Unfortunately, the law is not prepared to regulate careless speech. Existing speech regulation focuses on immediate harm; for example, slander and libel laws protect against reputational damage, while provisions against fraud protect against measurable financial harm based on deception. The cumulative, long-term risks of careless speech to degrade and homogenize knowledge, shared history and social truth is not something the law protects against. The rush to implement generative AI across journalism, education, science, healthcare and other sectors where the truth matters is thus very worrying.

Inaccurate statements and content can also be discriminatory.[120] Generative AI systems often perpetuate biases present in their training data, leading to outputs that may underrepresent or misrepresent protected groups (unequal representation) or even create new groups that the law does not protect.[121] This misalignment with established non-discrimination laws presents challenges, as traditional legal frameworks may not adequately address the nuances of algorithmic bias.[122] Moreover, when these systems generate abusive or derogatory content, it may qualify as harassment, broadening the scope of the application of existing non-discrimination statutes to AI-generated outputs.

Unfortunately, state-of-the-art tools to detect and mitigate bias are far from perfect. One of us has previously assessed the compatibility of current bias and fairness tests with EU and UK non-discrimination law. Of the most popular tests developed by the machine learning research community, 13 out of 20 do not live up to EU and UK legal standards.[123] The majority of existing tests have been developed in the US, where a different notion of fairness and discrimination prevails. Setting aside questions of legal alignment, enforcement of these tests to make AI models 'fairer' can also inadvertently harm people. Specifically, fairness is often achieved by 'leveling down', or making everyone worse off, rather than helping disadvantaged groups.[124]

There is some hope for improvement. As part of the implementation of the AI Act, the European Commission has authorized the creation of harmonized technical standards that should address the operational side of bias prevention. Of course, it is not clear whether these standards will actually address substantive social and ethical issues or merely create purely technical requirements for AI providers and users.[125]

3. Copyright Law

Copyright law is another significant factor influencing the development of generative AI.[126] Training datasets often include copyrighted material obtained through web scraping, raising concerns about intellectual property rights. The unauthorized use of such content for AI training has led to legal disputes, exemplified by cases like *The New York Times*' lawsuit against OpenAI and Microsoft,[127] and a landmark case in Germany concerning a photographer suing the association who had compiled the popular LAION-5B data set,[128] used by many generative AI applications. Copyright claims have both ways, with OpenAI accusing DeepSeek in January 2025 of illegally using their model outputs to train or 'distill' a rival model, which may be the first among many future legal battles between AI companies over the provenance of their generative AI models.[129]

Another key question for regulators to resolve in the future is whether the outputs of generative AI systems can be protected under copyright and intellectual property law. Current legal opinion in the EU and US trends toward outputs not being copyrightable in many cases, on the basis that content needs a human creator to receive protection.[130] Future cases may clarify what kind and degree of human involvement (e.g. tweaking and editing outputs) is necessary to justify protecting output by means of an exclusive right.[131] Remedies for copyright holders are far-reaching; hence, this field will continue to exercise significant impact in the development of generative AI, rivaling data protection law as the core regulatory substance.

The bottom line is this: while new regulatory initiatives, such as the EU's AI Act, are now freshly applicable, the current regulatory landscape for generative AI is predominantly shaped by existing laws. This may change once the high-risk rules of the AI Act fully kick in, starting in August 2026, and the prohibitions applicable since February 2025 are enforced. However, we anticipate that the 'old guard' of established legal frameworks will continue to play a major role in regulating generative AI.

V. The Future of Generative AI Regulation

This technology-neutral framework will prove particularly helpful in addressing future challenges, brought about by the increased integration of generative (and other types of) AI into layered or hybrid technological systems (see Section III.B.). It appears likely that going forward generative AI will be used less as a standalone technology, and more as a building block in increasingly complex and agentic decisioning systems across a range of applications in economic, military, and political contexts.

Indeed, recent advancements indicate a significant shift toward the integration of AI agents into various sectors.[132] Such agents' distinguishing characteristic is their capability to effectively tackle, in increasingly autonomous ways, specific tasks. A recent paper by Anthropic researchers 'draw[s] an important architectural distinction between workflows and agents: Workflows are systems where LLMs and tools are orchestrated through predefined code paths. Agents, on the other hand, are systems where LLMs dynamically direct their own processes and tool usage, maintaining control over how they accomplish tasks.'[133] They emphasize the development of effective agents by enhancing LLMs with capabilities such as retrieval, tool use, and memory. This approach is expected to enable AI systems to perform complex tasks with greater autonomy and adaptability.

Similarly, a much-noted Google white paper introduces AI agents designed to interact with external systems, process real-time data, and execute multi-step tasks independently.[134] These agents may automate functions traditionally managed by humans, changing and potentially enhancing business operations.[135] Echoing this sentiment, OpenAI CEO Sam Altman predicts that by 2025, AI agents will 'join the workforce' and materially influence company outputs.[136] In turn, in the academic realm, and related to the developments discussed in Section III.B, the 'Agent Laboratory' framework demonstrates the potential of LLM-based agents to conduct comprehensive research processes autonomously, or at least in human-machine teams.[137] (Partially) autonomous generative AI-based agents are, thus, not relegated to science fiction movies any more.

These applications are poised to have a significant impact on fields such as robotics,[138] but also potentially political discourse, and certainly the military realm, including the way in which data-driven processes are transforming warfare.[139] The battlefields of Ukraine and the Middle East testify to this development in many ways. The growing national security implications of AI, and frontier AI in particular, have already been recognized by many state and non-state actors. Executive Order 14578 on AI, enacted by US President Trump a mere three days after taking office in January 2025, is unapologetic about it, formulating as its sole policy goal to 'sustain and enhance America's global AI dominance',[140] expressly for societal, economic, and national security reasons.

Despite the remarkable technical progress, regulatory challenges may largely remain fairly similar to the ones in the last decade—but also as complex and unresolved as they were in these earlier phases of development. In our view, three aspects stand out.

First, a significant gap exists between the advanced capacities of AI systems and the scientific frameworks available for rigorous testing and evaluation.[141] Given the incentive structure of AI research and AI economics, where capability breakthroughs count more than technical breakthroughs, this gap may persist for quite some time. Evaluations continue to be empirically driven,[142] but struggle to keep pace with capability advancements. This discrepancy exacerbates the inherent time lags and uncertainties that accompany the governance of any emerging technology. This is particularly true for generative AI.

Second, precisely because of this uncertainty, the application of technology-neutral legal principles will remain crucial. Much progress has already been made within existing regulatory frameworks and institutions (as discussed in Section IV), which have demonstrated an ability to adapt to new challenges without the need for entirely novel legal systems.[143]

Third, however, the increasingly advanced, networked, and potentially agentic nature of frontier AI systems underscores the pressing need for a comprehensive international treaty on AI governance.[144] Such a treaty could streamline global efforts, establish consistent norms, and address the abuse potential of these systems, by state and non-state actors. Yet, the likelihood of achieving such an agreement remains highly uncertain given the current geopolitical climate marked by deepening tensions and competitive dynamics among major powers.

VI. Overview of the Handbook

Many of these, and more, topics are developed in much greater detail in the chapters of this volume. The book itself is organized in five larger parts, starting with this 'Introduction', followed by technical as well as socio-technical 'Foundations', then moving on to core 'Regulatory Challenges', such as generative AI's multifunctionality, inequality, 'hallucinations', liability for generative AI output, consumer law, and content moderation. Part IV then provides an account of generative AI 'Applications,' ranging from finance and medicine to agriculture. Finally, the section on 'Global and Regional Considerations' provides in-depth overviews of specific regions and countries concerning generative AI law and policy, and international law dimensions. The remainder of this chapter will lay out the core topics covered in the respective chapters, in the order of their appearance in the volume.

Following this introductory chapter, the next section ('Foundations') opens with a gentle technical introduction to generative AI and foundation models by Gerard de Melo and team.

Rumman Chowdhury lays the groundwork for responsible data governance in the era of generative AI and explores the organizational, legal, and technical dimensions of implementing responsible data practices. Her analysis takes into account ethical principles, evolving regulation, and technical strategies.

Gerard de Melo and team discuss the relevance of, and strategies for, generative AI prompting.

AI explainability, a concept that so far mainly has been developed and discussed for AI classification, may help build trust, identify biases and ascertain the behavior of an AI system is aligned with human reasoning. Wojciech Samek's chapter on 'Explaining and Interpreting Generative AI' discusses the different dimensions of how AI explainability can be brought to generative AI.

Agentic AI systems are characterized by their capacity to autonomously pursue goals and to adapt complex environments with limited supervisions. While potential benefits abound, key critical safety challenges range from unintended optimization to deceptive alignment and value drift, as Eleanor Watson discusses.

Benchmarks in turn, allow AI developers to study a model's safety and performance features. Generative AI's arrival has triggered significant interest and effort into both building—and understanding how to build—rigorous benchmarks for specialized domains. The chapter by Neel Guha, Julian Nyarko, Daniel E. Ho, and Christopher Ré Neel provides an overview of domain-specialized benchmarking and uses the legal field as a case study.

Detecting AI-generated content is crucial for maintaining information integrity, preventing misinformation, ensuring accountability, and upholding ethical standards; the undetected spread of AI-generated threatens to undermine democratic discourse. Gerard de Melo and team discuss challenges and opportunities for detecting AI-generated content.

The next two chapters move on to fundamental socio-technical dimensions of generative AI. As the emergence of new generative AI tools is beginning to reshape social relationships, Henry Shevlin explores the nascent field of 'Social AI', offering a framework for classifying human-AI relationships and their dynamics, discussing ethics at the frontier of human-AI relationships and addressing the more empirical matter of harms and benefits of human-AI relationships.

The section on foundations of generative AI concludes with Lynette Webb and Daniel Schönberger's examination of how generative AI has become central to existential risk debates. Their chapter highlights the key worries that underpin existential risk fears in relation to generative AI, as well as the key actions that governments and industry are taking thus far.

The book's next section, 'Regulatory Challenges', begins with Cary Coglianese and Colton R. Crum's analysis of how the heterogeneity and multifunctionality of foundation models and generative artificial intelligence complicates traditional regulatory approaches. The chapter argues that regulators will do well instead to promote proactive risk management on the part of developers and users by using management-based regulation, while regulators will also need to maintain ongoing vigilance and agility.

Roger Brownsword explores the application of the Rule of Law to generative AI. This chapter is in the nature of a prolegomenon to debates about the particular rules and principles that should be adopted for the governance of this latest technology.

Several jurisdictions are starting to regulate frontier AI systems (i.e. general-purpose AI systems that match or exceed the capabilities present in the most advanced systems). Jonas Schuett, Markus Anderljung, Alexis

Carlier, Leonie Koessler, and Ben Garfinkel, in their discussion of a regulatory approach for frontier AI, offer recommendations for transitioning from high-level principles to specific rules.

Providing access to open frontier AI models entails complexities that are explored by Seb Krier, Harry Law, and Kevin Klyman. Given the rapid pace of advancements in AI capabilities and commercialization, the chapter advocates for a nuanced approach to AI policy and open source.

Kelly Richdale discusses the application of generative AI in healthcare and pertinent regulatory challenges.

Ideally, Generative AI might come to play a role in addressing global inequality. Examining evidence from majority-world implementations in healthcare and education, Leah Junck, Fola Adeleke, and Rachel Adams showcase how generative AI can drive developmental priorities and discuss what safeguards are needed and available in low-resourced contexts to mitigate risk of harm to individuals, groups, and society.

Jerry John Kponyo, Francis Kemausuor, Eric Tutu Tchao, Henry Nunoo Mensah, and Rachel Yayra Adjoe explore the importance of promoting transparency in generative AI applications in key sectors such as agriculture, energy, and healthcare. By focusing on the African context, this chapter further aims to contribute to the global discourse on AI transparency.

As generative AI is increasingly used across sectors, its potential for societal benefit is paired with risks of discrimination. Philipp Hacker, Frederik Zuiderveen Borgesius, Brent Mittelstadt, and Sandra Wachter explore how generative AI challenges non-discrimination law. They examine the resources of existing EU law in addressing such challenges and offer suggestions on updating EU laws and mitigating biases preemptively.

Inspired by 'the five basic consumer rights' of the EU, Jan Trzaskowski and Marie Jull Sørensen frame generative AI in the context of consumer protection law, while drawing on EU digital regulation, including the GDPR, the AI Act and the Digital Services Act.

Jeremias Adams-Prassl examines the impact of generative AI on platform workers' rights, arguing for a focus on job quality rather than quantity and proposing regulatory responses to the challenges posed by algorithmic management in the gig economy.

The rise of Generative AI poses certain challenges to the current content moderation framework, as Pietro Ortolani argues in his investigation of the interplay between generative AI and content moderation. He advocates for an increased focus on the human-machine interaction and the use of generative AI to support humans in content moderation decision-making.

Generative AI can generate speech that could be problematic under a wide range of liability regimes. In particular, generative AI systems often generate outputs about real people, even when not explicitly prompted to do so, leading to significant reputational and privacy harms, especially when sensitive, misleading, or false. Reuben Binns and Lilian Edwards discuss what legal tools currently exist to protect such individuals, with a particular focus on defamation and data protection law.

Peter Henderson, Tatsunori Hashimoto, and Mark A. Lemley discuss three liability regimes under US law: defamation, speech integral to criminal conduct, and wrongful death. They argue that courts and policymakers should think carefully about what technical design incentives they create as they evaluate these issues.

An EU perspective is provided by Graziana Kastl-Riemann, who comprehensively examines liability for AI-generated speech under the German law of expression and offers observations from selected other EU states.

The impacts of generative AI on water, energy, carbon, waste, and land-use have recently gained considerable attention in the computer science community. Amy L. Stein encourages generative AI developers to integrate sustainability into their business growth strategies and provides suggestions for generative AI developers to achieve more sustainable generative AI practices.

Generative AI systems across modalities, ranging from text, code, image, audio, and video, have broad social impacts, but there is little agreement on which impacts to evaluate or how to evaluate them. Irene Solaiman, Zeerak Talat and a team of many more present a guide for evaluating base generative AI systems (i.e. systems without predetermined applications or deployment contexts).

Creativity and innovation are profoundly impacted by the rise of generative AI, which may assist human creators and inventors, or even replace them. Andreas Engel examines the intersection of copyright law and generative AI, focusing on both the training of AI systems and their outputs, also addressing international dimensions of the issue: international treaties, private international law rules, and extraterritoriality.

Herbert Zech discusses patent law from a European perspective, focusing on AI-related issues. He argues that while AI-specific legislative changes in patent law may not yet be necessary, the patentability of AI-generated output raises questions of inventorship.

Section IV ('Applications') demonstrates practical implementations of generative AI across sectors. Looking at the adoption of generative AI as a phenomenon at the intersection of economics and technology, Laurence Ales, Christophe Combemale, and Ramayya Krishnan discuss how from a firm perspective, generative AI differs from past technologies, and incentives for workers to adopt generative AI. Moreover, their chapter addresses open questions of AI and job displacement.

Alex P. Miller, Kartik Hosanagar, and Ramayya Krishnan explore the challenges and research opportunities associated with deploying LLMs for document and knowledge summarization in various business applications and for illustration, present a study on summarizing user-generated content from a product announcement video on social media.

Based on his involvement in pioneering work in the field, Alexandre Zavaglia Coelho explores the use of AI techniques, including generative AI, in the legal fields, encompassing both the public and private sectors, with a particular focus on AI's capacity to utilize underexplored resources, such as legal documents.

International arbitration is an area of practice particularly suitable for examining generative AI's impact, owing to unique characteristics such as its multijurisdictional and multilingual scope. Elizabeth Chan, Kiran Nasir Gora and Eliza Jiang explore how generative AI tools, such as ChatGPT, Harvey AI, and Co-Counsel, are reshaping international arbitration practice.

Joyce Nakatumba-Nabende, Ann Lisa Nabiryo, Peter Nabende, Jennifer Winfred Namuyanja, Andrew Katumba, and Derrick Ssekidde discuss how generative AI may bridge information gaps that smallholder farmers face by means of LLM-powered question-answer systems: their chapter describes the process of curating and evaluating a crop and animal question-answer dataset and analyses the potential of LLMs to provide context-specific agricultural advisory to farmer questions.

Education plays a crucial role in enabling critical engagement with AI, as Julia Powles emphasizes. In her chapter, she presents a practical, extensible, novel guide to AI for learners and educators at all stages, the POWER (Power, Oversight, Workers, Environment, Redress) guide to AI, and applies it to the educational realm.

Sarah Hammer examines AI's transformative impact on the financial sector and its regulatory challenges. The chapter provides a taxonomy for explainability in financial services and argues that while explainability provides a foundation for AI policymaking in finance, a multifaceted approach balancing innovation with risk management is necessary.

Cybersecurity represents a significant risk in the adoption of generative AI. Thomas Wischmeyer and Michael Strecker delineate the specific security threats posed by generative AI and explore potential regulatory responses, underscoring the need to strike a balance between fostering innovation and implementing robust cybersecurity measures to prevent the misuse of advanced generative AI systems.

Section V ('Global and Regional Considerations') introduces regulatory approaches across jurisdictions. Jeannie Marie Paterson examines Australia's efforts to design 'guardrails' that ensure safety without stifling innovation. Her work highlights the importance of aligning national regulatory strategies with international standards, fostering coherence, and avoiding undue complexity.

Dora Kaufman analyzes Brazil's hybrid approach which combines American, British, and European elements and provides for a hybrid structure that values existing sectoral regulatory agencies as members of a national regulation and governance system (SIA) with a coordinating authority while addressing some of the country's structural challenges.

The US and Canada regulate generative AI in different ways for the public and private sectors, as Ignacio Cofone discusses in his chapter. In addition, both countries have federal and local legal frameworks that are relevant to and apply to generative AI.

Rogier Creemers demonstrates how China's swift regulatory response to the rapid development of AI applications seeks to balance political considerations. This includes addressing concerns over politically undesirable and destabilizing content on the one hand, and China's bid to attain global AI leadership on the other.

The EU's adaptation of its AI Act to address foundation models is traced by Kai Zenner, who explores the legal challenges of that exercise, as well as the evolving regulatory digital landscape in the EU, aimed at balancing innovation with safety and fairness.

India's unique position between technical capability and social complexity leads to opportunities and risks for marginalized populations, as Debayan Gupta explains. His chapter highlights how proper regulation will facilitate generative AI's potential to drive inclusive growth while cautioning against the deepening of existing inequalities.

Adrian Ang and Alexander Yap examine Singapore's national AI strategy, which was released in 2019, and subsequent implementing steps, including changes to copyright and data protection laws, and the lifting of a data center moratorium. Their analysis also addresses content regulation and new anti–fake news legislation.

Christopher T. Marsden analyzes generative AI regulation in the UK as a test case for the reassessment of the government's approach to technology regulation: its approach has been to 'first, do no harm' except in cases of disinformation and deepfakes generated by AI. The chapter argues that the UK government's 'unregulation' of AI has been aptly termed 'regulatory hallucination'.

As solving many of the challenges presented by AI requires international coordination and cooperation, the volume concludes with Matthijs Maas and José Villalobos advocating for a Framework Convention on AI. They propose a hybrid model combining broad scope with specific obligations and implementation mechanisms concerning issues on which states already converge.

Notes

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11 See e.g. ibid, 5.

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